KAIZEN MATHS

Lesson starter

Factorising Quadratic Expressions

Monic Quadratics

Name _____

S1

Factorise each expression.

Factorise: $x^2 + 8x + 15$

S2

Factorise each expression.

Factorise: $x^2 - 6x + 8$

S3

Factorise each expression.

Factorise: $x^2 - 36$

S4

Factorise each expression.

Factorise: $x^2 + 13x + 42$

Show your working

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ALG-CFE2-L1 · S

Show your working

KAIZEN MATHS

Exit ticket

Factorising Quadratic Expressions

Monic Quadratics

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E1

Factorise each expression.

Factorise: $x^2 - x - 56$

E2

Factorise each expression.

Factorise: $16x^2 - 49$

E3

Factorise each expression.

Factorise: $x^2 + 12x + 35$

ALG-CFE2-L1 · E

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ALG-CFE2-L1 · E

Show your working

Teacher answers & working

Factorising Quadratic Expressions · Monic Quadratics

ALG-CFE2-L1 · Student pages 1-2. Print at 100%, single-sided. Cut on dashed lines.

S1

Factorise each expression.

Factorise: $x^2 + 8x + 15$

Answer:

$(x + 5)(x + 3)$

Working:

Find two numbers whose product is the constant term and whose sum is the coefficient of x.

$5 \times 3 = 15 \quad \text{and} \quad 5 + 3 = 8$

$x^2 + 8x + 15$

$= x^2 + 5x + 3x + 15$

$= x(x + 5) + 3(x + 5)$

$= (x + 5)(x + 3)$

Check by expanding the brackets.

S2

Factorise each expression.

Factorise: $x^2 - 6x + 8$

Answer:

$(x - 2)(x - 4)$

Working:

Find two numbers whose product is the constant term and whose sum is the coefficient of x.

$-2 \times -4 = 8 \quad \text{and} \quad -2 + -4 = -6$

$x^2 - 6x + 8$

$= x^2 - 2x - 4x + 8$

$= x(x - 2) - 4(x - 2)$

$= (x - 2)(x - 4)$

Check by expanding the brackets.

Teacher answers & working

Factorising Quadratic Expressions · Monic Quadratics

ALG-CFE2-L1 · Student pages 1-2. Print at 100%, single-sided. Cut on dashed lines.

S3

Factorise each expression.

Factorise: $x^2 - 36$

Answer:

$(x - 6)(x + 6)$

Working:

Recognise the difference of two squares pattern.

$A^2 - B^2 = (A - B)(A + B)$

$A = x, \quad B = 6$

$A - B = x - 6$

$A + B = x + 6$

$x^2 - 36 = (x - 6)(x + 6)$

Check by expanding the conjugate brackets.

S4

Factorise each expression.

Factorise: $x^2 + 13x + 42$

Answer:

$(x + 7)(x + 6)$

Working:

Find two numbers whose product is the constant term and whose sum is the coefficient of x.

$7 \times 6 = 42 \quad \text{and} \quad 7 + 6 = 13$

$x^2 + 13x + 42$

$= x^2 + 7x + 6x + 42$

$= x(x + 7) + 6(x + 7)$

$= (x + 7)(x + 6)$

Check by expanding the brackets.

Teacher answers & working

Factorising Quadratic Expressions · Monic Quadratics

ALG-CFE2-L1 · Student pages 1-2. Print at 100%, single-sided. Cut on dashed lines.

E1

Factorise each expression.

Factorise: $x^2 - x - 56$

Answer:

$(x + 7)(x - 8)$

Working:

Find two numbers whose product is the constant term and whose sum is the coefficient of x.

$7 \times -8 = -56 \quad \text{and} \quad 7 + -8 = -1$

$x^2 - x - 56$

$= x^2 + 7x - 8x - 56$

$= x(x + 7) - 8(x + 7)$

$= (x + 7)(x - 8)$

Check by expanding the brackets.

E2

Factorise each expression.

Factorise: $16x^2 - 49$

Answer:

$(4x - 7)(4x + 7)$

Working:

Recognise the difference of two squares pattern.

$A^2 - B^2 = (A - B)(A + B)$

$A = 4x, \quad B = 7$

$A - B = 4x - 7$

$A + B = 4x + 7$

$16x^2 - 49 = (4x - 7)(4x + 7)$

Check by expanding the conjugate brackets.

Teacher answers & working

Factorising Quadratic Expressions · Monic Quadratics

ALG-CFE2-L1 · Student pages 1-2. Print at 100%, single-sided. Cut on dashed lines.

E3

Factorise each expression.

Factorise: $x^2 + 12x + 35$

Answer:

$(x + 5)(x + 7)$

Working:

Find two numbers whose product is the constant term and whose sum is the coefficient of x.

$5 \times 7 = 35 \quad \text{and} \quad 5 + 7 = 12$

$x^2 + 12x + 35$

$= x^2 + 5x + 7x + 35$

$= x(x + 5) + 7(x + 5)$

$= (x + 5)(x + 7)$

Check by expanding the brackets.